



ÚRAD PODPREDESEDU VLÁDY
SLOVENSKEJ REPUBLIKY
PRE PLÁN OBNOVY
A ZNALOSTNÚ EKONOMIKU

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WP3 — DEVELOPMENT AND IMPLEMENTATION OF THE TRAINING STRATEGY



Solution options: distribution comparison and index usage

Seminar - implementation variants



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Context and aim

The WP3 seminar opens the solution space for the upcoming implementation (the D12 code). Instead of a single approach, we present and compare several options.



Distribution comparison — Which statistical measures to use for comparing the weight distribution.



Index usage — How to turn a measure into a usable signal (granularity, role, decision-making).



Quantisation granularity — Per-tensor vs per-channel — what the index actually sees.



Recommended direction — Selecting a combination for D12 while keeping extensions open.

The core task

We need to compare the weight distribution of the **reference (FP32)** and **quantised** model and turn it into a usable signal - the index.



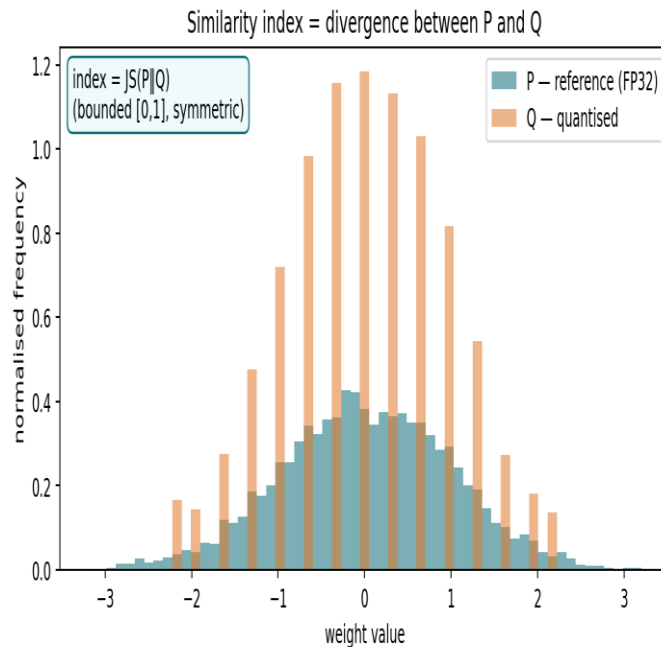
How to compare distributions?

Choice of statistical measure of difference.



How to use the index?

Granularity, role in training, decision-making.



Methods for comparing distributions



KL divergence

Asymmetric, sensitive to tails; measures information loss.



JS divergence

Symmetric, bounded $[0,1]$; suited to decision-making.



Cosine similarity

Compares the direction of weight vectors; very cheap.



Wasserstein distance

“Earth-mover”; handles even non-overlapping distributions.

Comparison of metrics

Metric	Bounded	Symmetric	Cost	Suited to
KL divergence	no	no	low	detecting information loss
JS divergence	yes [0,1]	yes	low	thresholding and automation
Cosine similarity	yes [-1,1]	yes	very low	rough comparison of direction
Wasserstein dist.	no	yes	higher	geometric differences

How to use the index — variants



Granularity

global (whole model)

vs

per-layer (most sensitive layer)



Role in training

predictor (pre-training — bit-width selection)

vs

monitor (during QAT — per epoch)



Reference

frozen FP32 reference

vs

epoch-over-epoch (inter-epoch change)



Decision-making

fixed threshold $JS \leq \tau$ (bit-width selection)

vs

+ early stopping at stabilisation

Per-tensor vs per-channel

The choice affects what the index sees and how quantisation degrades deep networks.



Per-tensor

one scale for the whole kernel

- + simple and fast
- + fewer scaling parameters
- fails for deep networks with weight outliers (accuracy collapse)



Per-channel

own scale per output channel

- + robust even at lower bit-widths
- + near-lossless for deep networks
- slightly higher computational cost

Recommended direction for D12



Primary metric : JS divergence (bounded, thresholdable); KL and cosine as complementary.



Index granularity : both global and per-layer — overall state and the most sensitive layer.



Role : both predictor and monitor



Quantisation : per-channel - robustness for deep networks.



Characterisation : moments + entropy + histograms with consistent bin edges.

Next steps and discussion



Implementation of the chosen direction as the D12,D13 code.



Validation on datasets and architectures (WP4).

Discussion

Thank you for your attention.

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